AI Memory Forensic Investigation Assistant

INVESTIGATION REPORT

Investigation ID	INV-20260805-48E690
Case Name	INV-20260805-48E690
Generated	2026-08-05 12:51:05
Memory Dump Filename	tmp5vwg6s93.raw
SHA-256	9bf29047efa9eb01852d34770f3c443e16908664be77acb176623051f0cf40a1
Dump Size	49 B

Investigation Summary

Investigation Status	completed
Total Plugins Executed	10
Successful Plugins	4
Failed Plugins	6
Total Evidence	10
Investigation Duration	0.3s

This report summarizes the forensic analysis performed on the memory dump for investigation INV-20260805-48E690. Plugin execution results, parsed evidence, high-priority findings, and AI-assisted question answering are documented in the sections below.

Plugin Execution Summary

#	Plugin	Status	Execution Time
1	windows.malfind	failed	0.00s
2	windows.registry.printkey	failed	0.00s
3	windows.filescan	completed	0.00s
4	windows.netscan	failed	0.00s
5	windows.handles	failed	0.00s
6	windows.dlllist	failed	0.00s
7	windows.cmdline	completed	0.00s
8	windows.pstree	failed	0.00s
9	windows.pslist	completed	0.00s
10	windows.info	completed	0.00s

6 plugin(s) failed to execute. See the application logs for details.

Evidence Summary

Evidence grouped by artifact type:

Artifact Type	Count	Confidence
cmdline	3	100%
filescan	2	100%
info	2	100%
pslist	3	100%

Artifact breakdown:

cmdline

windows.cmdline (confidence 100%): {"pid": 4321, "cmd": "cmd.exe /c whoami"}
windows.cmdline (confidence 100%): {"pid": 5555, "cmd": "C:\\Temp\\malware.exe --connect 10.0.0.5:4444"}
windows.cmdline (confidence 100%): {"pid": 1234, "cmd": "explorer.exe"}

filescan

windows.filescan (confidence 100%): {"offset": "0x1", "name": "C:\\Temp\\malware.exe", "size": 123456}
windows.filescan (confidence 100%): {"offset": "0x2", "name": "C:\\Windows\\System32\\svchost.exe", "size": 50000}

info

windows.info (confidence 100%): {"name": "System", "pid": 4, "state": "running"}
windows.info (confidence 100%): {"name": "explorer.exe", "pid": 1234, "state": "running"}

pslist

windows.pslist (confidence 100%): {"pid": 1234, "name": "explorer.exe", "path": "C:\\Windows"}
windows.pslist (confidence 100%): {"pid": 5555, "name": "malware.exe", "path": "C:\\Temp\\malware"}
windows.pslist (confidence 100%): {"pid": 4321, "name": "cmd.exe", "path": "C:\\Windows\\System32"}

High Priority Findings

High Confidence Evidence

Plugin	Artifact	Confidence	Value
windows.info	windows.info	100%	{"name": "System", "pid": 4, "state": "running"}
windows.info	windows.info	100%	{"name": "explorer.exe", "pid": 1234, "state": "running"}
windows.pslist	windows.pslist	100%	{"pid": 1234, "name": "explorer.exe", "path": "C:\\Windows"}
windows.pslist	windows.pslist	100%	{"pid": 5555, "name": "malware.exe", "path": "C:\\Temp\\malware"}
windows.pslist	windows.pslist	100%	{"pid": 4321, "name": "cmd.exe", "path": "C:\\Windows\\System32"}
windows.cmdline	windows.cmdline	100%	{"pid": 4321, "cmd": "cmd.exe /c whoami"}
windows.cmdline	windows.cmdline	100%	{"pid": 5555, "cmd": "C:\\Temp\\malware.exe --connect 10.0.0.5:44
windows.cmdline	windows.cmdline	100%	{"pid": 1234, "cmd": "explorer.exe"}
windows.filescan	windows.filescan	100%	{"offset": "0x1", "name": "C:\\Temp\\malware.exe", "size": 123456}
windows.filescan	windows.filescan	100%	{"offset": "0x2", "name": "C:\\Windows\\System32\\svchost.exe", "s

Malware Findings

Plugin	Artifact	Confidence	Value
windows.pslist	windows.pslist	100%	{"pid": 5555, "name": "malware.exe", "path": "C:\\Temp\\malware"}
windows.cmdline	windows.cmdline	100%	{"pid": 5555, "cmd": "C:\\Temp\\malware.exe --connect 10.0.0.5:44
windows.filescan	windows.filescan	100%	{"offset": "0x1", "name": "C:\\Temp\\malware.exe", "size": 123456}

Network Findings

Plugin	Artifact	Confidence	Value
windows.cmdline	windows.cmdline	100%	{"pid": 5555, "cmd": "C:\\Temp\\malware.exe --connect 10.0.0.5:44

Suspicious Artifacts

None identified.

AI Investigation Summary

The investigator asked the AI assistant questions about the evidence. Each answer is grounded in retrieved forensic evidence and cites the supporting records.

Q1. Which processes ran suspicious commands?

Answer: Based on [1] and [2], the process cmd.exe was observed running suspicious commands with an elevated confidence.

Evidence citations:

- windows.pslist [confidence 100%]
- windows.cmdline [confidence 100%]

Q2. Which processes ran suspicious commands?

Answer: Based on [1] and [2], the process cmd.exe was observed running suspicious commands with an elevated confidence.

Evidence citations:

- windows.pslist [confidence 100%]
- windows.cmdline [confidence 100%]

Q3. Which processes ran suspicious commands?

Answer: Based on [1] and [2], the process cmd.exe was observed running suspicious commands with an elevated confidence.

Evidence citations:

- windows.pslist [confidence 100%]
- windows.cmdline [confidence 100%]

Q4. Which processes ran suspicious commands?

Answer: Based on [1] and [2], the process cmd.exe was observed running suspicious commands with an elevated confidence.

Evidence citations:

- windows.pslist [confidence 100%]
- windows.cmdline [confidence 100%]

Q5. describe this investigation

Answer: The investigation reveals the presence of a malicious process (malware.exe, PID 5555) executing from the file system path "C:\\Temp\\malware" [3], which was launched with the command-line argument "--connect 10.0.0.5:4444" [2]. This suggests the process is attempting to establish a network connection to the IP address 10.0.0.5 on port 4444. The file "malware.exe" exists in the "C:\\Temp\\" directory, as confirmed by the file scan evidence [5]. Additionally, two legitimate processes are noted: the System process (PID 4) [1] and explorer.exe (PID 1234) [4], both in a "running" state.

Evidence citations:

- windows.pslist [confidence 100%]
- windows.cmdline [confidence 100%]
- windows.filescan [confidence 100%]
- windows.info [confidence 100%]
- windows.info [confidence 100%]

Q6. What processes are running suspicious commands?

Answer: The process running a suspicious command is "malware.exe" with PID 5555, which executed the command `--connect 10.0.0.5:4444` [2]. This indicates a potential network connection to a remote server, which is often associated with malicious activity. The other process, "cmd.exe" (PID 4321), executed the benign command `whoami` [3], which does not indicate suspicious behavior based on the provided evidence.

Evidence citations:

- windows.cmdline [confidence 100%]
- windows.cmdline [confidence 100%]

Conclusion

The investigation of memory dump tmp5vwg6s93.raw (investigation INV-20260805-48E690) executed 10 Volatility plugin(s) (4 successful, 6 failed) and produced 10 evidence record(s).

The analysis identified 14 high-priority finding(s) including high-confidence evidence, malware indicators, network activity, and suspicious artifacts. These findings should be reviewed manually by a forensic analyst before reaching a final conclusion.

The AI investigation assistant answered 6 question(s) during this investigation. Where evidence was insufficient, the assistant explicitly stated that the answer could not be determined from the available evidence.

It is recommended to treat the findings in this report as indicative rather than conclusive. Corroborate each finding with additional analysis tools and preserve the original memory dump for chain of custody.

Report generated by the AI Memory Forensic Investigation Assistant on 2026-08-05 12:51:05.